black hole tears a neutron star apart, when the universe was born, and in many other ways.

One day in 2019, LIGO was hit by a burst of gravitational waves far stronger than any ever before seen (Figure 16.2). The waves oscillated with an amplitude that grew and fell several times, and then cut off suddenly. The entire burst lasted for only a few seconds.

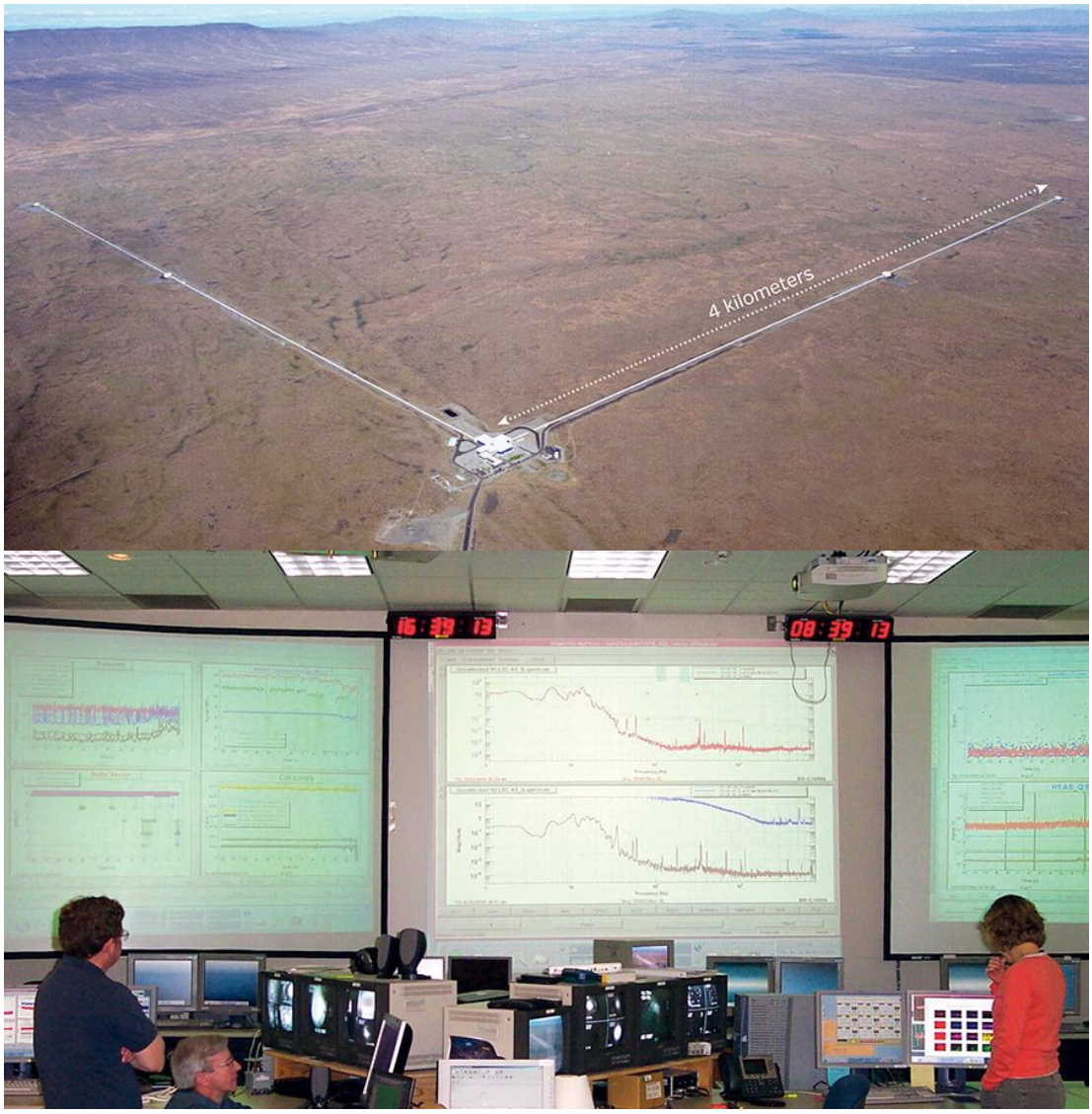


Fig. 16.1. *Top*: Aerial photograph of the LIGO gravitational wave detector at Hanford, Washington. *Bottom*: The LIGO control room where the detector is controlled and its signals are monitored.